Nowcasting Pertussis Case Counts in Washington State

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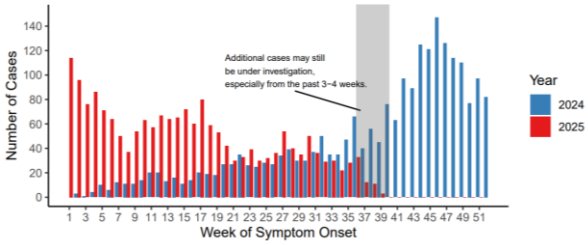
BACKGROUND

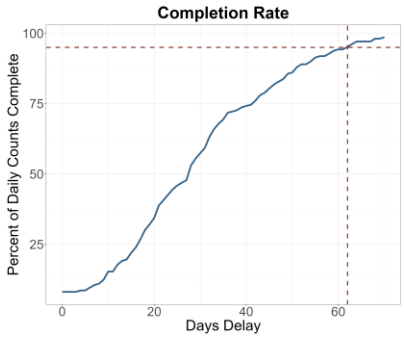
Pertussis (whooping cough) is a highly contagious respiratory infection known for its prolonged illness (the "100-day cough"). Beginning in 2024, Washington state experienced a significant rise in cases. Delays in diagnosis, testing, and case investigation result in cases taking weeks to be fully reported, making it appear as though cases are declining, regardless of the true trend. Nowcasting makes use of the delay patterns to estimate the final counts from the incomplete data, helping us better understand the true trend in case counts.

Figure 1: Number of Pertussis Cases Reported in Washington State by CDC Week of Symptom Onset: 2024 (blue) vs 2025 (red)



DATA CHALLENGES

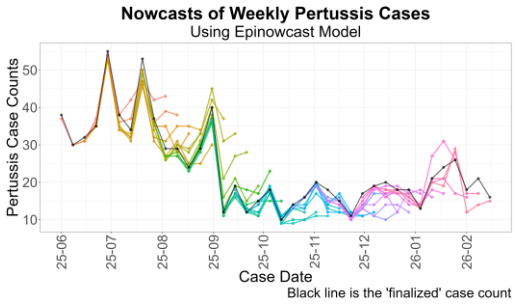
- Long reporting delays (up to 9 weeks)
- The preferred event date is the *date of symptom onset*, which is not initially reported, but is determined through case investigations
- Case counts for a given day can decrease as the event date is updated
- The existing database does not track updates to the event date, preventing retrospective observation of the complete data delay



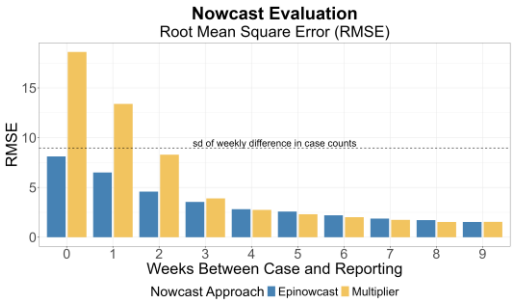
METHODS

- Implemented a process to track data changes
- Epinowcast[1] Bayesian framework for nowcasting
    - Incorporated data without a symptom onset data as 'missing data'
- Compared to a simple multiplier approach, where reported counts were scaled up by historical reporting proportions

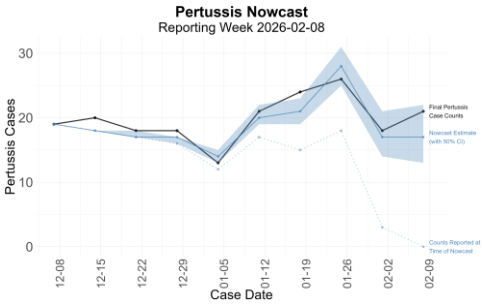
RESULTS



Nowcast estimates closely follow final case counts, though early nowcasts overestimate cases during periods of rapid decline



The Epinowcast estimate outperformed a simple multiplier approach at early time points, producing reasonable estimates by a 2-week delay



Comparison of nowcast, currently reported cases, and final reported counts produced February 8th, 2026

Nowcasting enables earlier insight into pertussis trends by shortening the data delay from up to 9 weeks to within 2 weeks for reasonable case count estimates

DISCUSSION We are working to incorporate nowcasting into our reporting pipeline to improve real-time understanding of pertussis trends. Ongoing monitoring and evaluation will be necessary to assess accuracy and ensure estimates remain reliable during larger outbreaks.

REFERENCES 1. Sam Abbott, Lison A, Funk S, Pearson C, Gruson H, Guenther F, DeWitt M, Azam J, Sebastian J (2026). _epinowcast: A Bayesian Framework for Real-time Infectious Disease Surveillance_. doi:10.5281/zenodo.5637165 <https://doi.org/10.5281/zenodo.5637165>.